

Experiment- 4

AC Sonometer

AIM:

To find the frequency of the AC mains with a sonometer.

APPARATUS:

A sonometer (with soft iron wire), ½ kg hanger, seven ½ kg slotted weights, an electromagnet with a step-down transformer, clamp stand, rubber pad, meter scale.

THEORY:

Frequency of AC mains using a sonometer

$$v = \frac{1}{4\sqrt{m}} \frac{\sqrt{T}}{l}$$

Where l is the resonating length, T is the tension in the string, and m is the mass per unit length of the wire.

PROCEDURE:

1. Place the sonometer on the table such that the pulley projects beyond the edge of the table. Put suitable weights on the hanger.
2. Move the bridges to the outward position so that maximum length can be accommodated.
3. Clamp the electromagnet in a stand, keeping one of its cores very close to the middle of the sonometer wire.
4. Switch on the electromagnet and adjust the length of the vibrating wire by moving the knife edges with small length until the wire begins to vibrate.
5. Now make minor adjustments to make the amplitude of the vibrating wire maximum; in that case, we will hear a hissing sound in the wire.
6. Note the load on the wire and measure the length of the vibrating position of the wire in both length-increasing and length-decreasing modes; i.e., l_1 and l_2
7. Now change the load of the hanger and repeat the same procedure to obtain resonating length of the wire.
8. Take 5 sets of such readings.

RESULT:

The frequency of alternating current=

Precautions

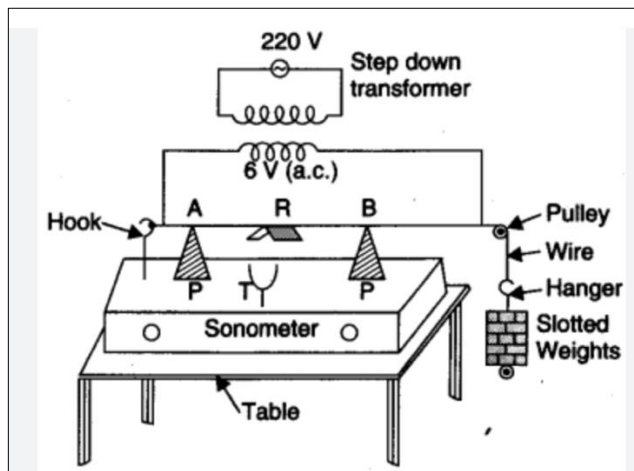
- i. The wire should be of soft iron or of any other magnetic material.
- ii. The tip of the electromagnet should be very close to the wire in its middle.

Sources of Error

- i. Wire may not be rigid and of uniform cross-sectional area.
- ii. The pulley may not be frictionless.

Diagram and observation table on the left side of the page

Sonometer Diagram:



Observation table

Linear density of the wire, $m =$

S.No.	Tension $T = mg$ (N)	Resonant length			$\frac{\sqrt{T}}{l}$
		Length (l_1) increasing	Length (l_2) decreasing	Mean $l = \frac{l_1 + l_2}{2}$	$(\frac{\sqrt{N}}{m})$
1					
2					
3					
4					

Calculations:

Mean $= \frac{\sqrt{T}}{l} =$

$$v = \frac{1}{4\sqrt{m}} \frac{\sqrt{T}}{l} =$$